

THE OPEN UNIVERSITY OF SRI LANKA
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING
BACHELOR OF SOFTWARE ENGINEERING
EEI6373 – Performance Modelling
Mini Project- Deliverables

Performance Modeling of a Webhook Delivery System Under Increasing
Workload and Worker Scalability

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1. System Details

The system considered in this mini project is a webhook delivery system developed in a software engineering project. The system receives events from clients and delivers those events to external webhook endpoints.

The system uses an asynchronous processing architecture. When an event is received through the FastAPI application, the event is persisted in PostgreSQL and a delivery job is placed in a Redis-based queue. RQ workers retrieve jobs from the queue and perform HTTP requests to the target webhook endpoint. This separates event reception from webhook delivery and allows delivery work to be processed asynchronously.

The performance study focuses on the core queue and worker-processing path. The later Kubernetes deployment work is outside the scope of this performance model.

2. High Level Problem

As the number of incoming webhook events increases, the available processing capacity of the worker may become insufficient to process events at the same rate at which they arrive. When this occurs, events spend more time waiting in the queue before being processed. This can increase delivery latency and reduce the system's ability to respond efficiently to higher workloads.

The implemented system uses a single worker, which provides a useful baseline for investigating this problem. A performance model can then be used to evaluate alternative worker configurations without physically deploying every possible configuration.

The main problem investigated in this study is therefore the relationship between incoming workload and available worker capacity. The study examines how increasing the workload affects system performance and how increasing the number of workers affects the system's ability to handle that workload.

3. Dataset

The dataset for this study is generated using a controlled discrete-event simulation of the webhook delivery system. The deployed system was not tested using a planned range of workloads and worker configurations, so its available monitoring data does not provide a suitable controlled dataset for comparing all of the conditions required by this study. The simulation therefore provides a consistent way to test different arrival rates and worker configurations.

The simulation represents webhook events arriving over time, waiting when all workers are busy, and being processed when a worker becomes available. The model uses an M/M/c queueing representation, where the arrival rate is represented by λ , the service rate of one worker by μ , and the number of workers by c . The service rate used in the model is an explicit modelling assumption rather than a measurement from the deployed system.

3.1 Dataset Variables

Variable	Type	Description
Arrival rate (λ)	Independent	Number of webhook events arriving per second.
Number of workers (c)	Independent	Number of workers available to process events.
Processing latency	Dependent	Time required for an event to pass through the simulated processing path.
Throughput	Dependent	Number of events successfully processed per second.
Queue length	Dependent	Number of events waiting for processing.
Worker utilization	Dependent	Percentage of available worker capacity being used.

3.2 Dataset and Experimental Conditions

Three controlled experiments are used to generate and compare the dataset. In the first experiment, the workload is increased while keeping one worker. In the second experiment, the arrival rate is fixed at 8 events per second while the number of workers is increased from one to four. In the third experiment, workload and worker capacity are increased together, with the workload increasing from 8 to 32 events per second and the worker count increasing from one to four.

The simulation records latency, throughput, mean queue length, and worker utilization for each tested configuration. The resulting observations are used to compare system behaviour under different workload and processing-capacity conditions.

4. Performance Objectives

The main objective of the study is to evaluate how the performance of the webhook delivery system changes as incoming workload increases and as the number of workers is changed.

Performance objective	Purpose
Latency	Measure how the time required to process a webhook event changes as workload increases and workers become more heavily loaded.
Throughput	Measure the rate at which webhook events can be successfully processed and determine how

	much workload the system can handle under different worker configurations.
Queue Length	Examine how many events are waiting for processing and identify conditions where incoming workload begins to exceed available processing capacity.
Worker Utilization and Scalability	Evaluate how heavily the available workers are being used and whether increasing the number of workers provides meaningful performance improvements.

Latency is used as the primary metric for visual comparison because it directly shows the effect of workload and worker capacity on the time required to process an event. Throughput, queue length, and worker utilization are used as supporting measures to explain the changes observed in latency and to identify conditions where the system approaches its processing capacity.

5. Performance Question

How does increasing workload affect the performance of the webhook delivery system, and how does increasing the number of workers affect its ability to handle that workload?

6. Scope

The study concentrates on the queue and worker-processing stage of the webhook delivery system. The purpose is to understand workload, queueing, latency, throughput, and worker capacity under controlled conditions. The results are simulation results and are not presented as direct measurements of the deployed production system.

7. Project Repository

The supporting materials for this mini project are available in the performance-modeling branch of the repository.

GitHub repository: <https://github.com/Vimukthi-Randunu/webhook-delivery-system>